

Modeling and Compensation for Angular Transmission Error of Harmonic Drive Gearing in High Precision Positioning

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Abstract— This paper presents a novel modeling and compensation approach for the angular transmission error in harmonic drive gearing. In the modeling, physical phenomena of the transmission error due to nonlinear elastic deformations in micro-displacement region are especially dealt with, as well as the synchronous component which has been discussed in a variety of conventional studies. Based on the analyses of the phenomena, the nonlinear elastic component is mathematically modeled by applying a modeling framework for the rolling friction with hysteresis attributes. The proposed transmission error model has been adopted to the positioning system as a model-based feedforward compensation manner. Experimental results using a prototype show the effectiveness of the proposed modeling and compensation.

I. INTRODUCTION

Harmonic drive gearing is widely used in a variety of industrial applications, e.g. industrial and/or humanoid robots, precision positioning devices, etc., because of their unique kinematics and high performance attributes such as simple and compact mechanism, high gear ratios, high torque, and zero backlash. As shown schematically in Fig.1, this transmission system is generally comprised of just three components: a wave generator (WG) with an elliptical shape, a flexspline (FS) of an elastic thin-walled steel cup, and a circular spline (CS) of a rigid steel ring with internal teeth, which was developed to take advantages of the elastic dynamics of metal. The basic principles of motion are as follows: FS is deflected by WG into an elliptical shape and the inscribed ellipse contacts with internal teeth of CS at two points. Then, the tooth engagement position moves by turns relative to CS, while FS moves by two teeth relative to CS because of two fewer teeth of FS than that of CS. As a result, the high gear ratio can be achieved without complex mechanisms and structures. Details of its operation can be found in [1].

Typical mechatronic devices including harmonic drive gearing, such as industrial robots, are generally controlled in a semi-closed control manner: actuator states (displacement and/or velocity) can be directly detected and controlled by sensors on the actuator shaft, while load states are indirectly controlled with the apparent sensor resolution due to the gear ratio. However, since the transmission systems inherently possess nonlinear attributes known as “angular transmission

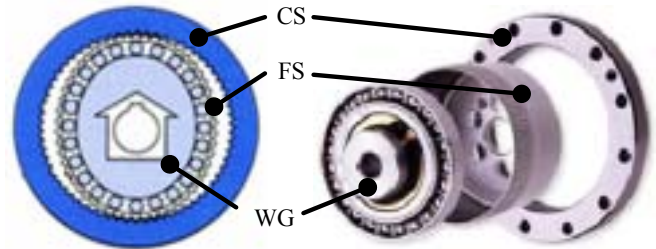


Fig. 1. Components of harmonic drive gearing.

errors” due to the structural error and flexibility, the ideal control accuracy corresponding to the apparent resolution can not be attained in the load side. Although the harmonic drive gearing can substantially reduce the amount of the errors because of their zero backlash, the angular transmission errors due to kinematic errors and nonlinear elastic deformations in FS are still remained, resulting in the deterioration in the control performance [2], [3], [4].

In this research, the angular transmission errors are classified into two: One is “synchronous component” of a periodical angular transmission error due to the structural errors in manufacturing process of gears, which is synchronous to the rotation of gearing. The other is “nonlinear elastic component” which behaves as a nonlinear phenomenon with hysteresis due to the nonlinear elastic deformations. The periodical error of synchronous component has been widely analyzed and mathematically modeled in various conventional studies [5], [6]. Since the synchronous component can be generally modeled as a periodical pulsation for the rotational angle of gearing, the error can be compensated in a model-based feedforward manner. The nonlinear elastic component, on the other hand, has been usually handled as an elastic element in mechanism, e.g. a low stiffness at robot joint, which caused torsional vibrations (mechanical resonant vibrations) between actuator and load during motion transient [7]. A variety of vibration suppression control approaches, therefore, have been widely applied to provide higher control performance and/or robustness against plant perturbations [8], [9]. The nonlinear elastic attribute, in addition, may deteriorate the static control accuracy due to the essential transmission error with hysteresis, especially in micro-displacement region, e.g. several tens to hundred arc-sec ($1 \text{ arc-sec} = 1/3600 \text{ degrees}$) in load angle. For example, at the settling in fine positioning, this transmission error

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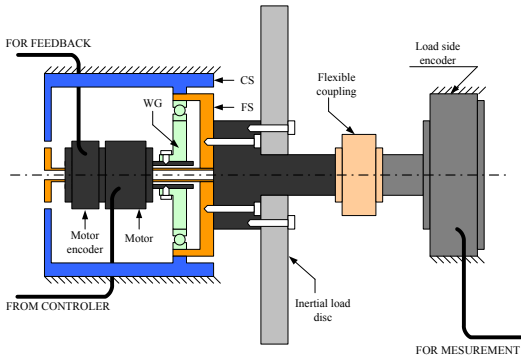


Fig. 2. Experimental setup.

causes the steady state angular error in load side, while the actuator angle can be settled at the exact target position. This distinctive static phenomenon has not yet been discussed well, unlike the dynamic behaviors leading to mechanical vibrations.

Under the background above, this paper presents a novel precise modeling methodology for the angular transmission error to improve the static positioning accuracy [10], where the nonlinear elastic component in micro-displacement region is mathematically modeled by applying a modeling framework for the rolling friction with hysteresis attributes [11], [12], as well as the conventional modeling for the synchronous component by spectrum analyses for rotation angle [4]. This transmission error model is then adopted to the positioning system as a model-based feedforward compensation manner, in order to improve the settling accuracy. The proposed modeling and compensation have been verified by a series of numerical simulations and experiments using a laboratory prototype.

II. CONFIGURATION OF EXPERIMENTAL SETUP

Fig.2 shows a schematic configuration of the laboratory prototype as an experimental positioning device, which is comprised of an actuator (AC motor) with an encoder, a harmonic drive gearing, an inertial load, and a load side encoder. Specifications of the prototype are listed in Table I. This positioning device is controlled in a typical semi-closed control manner by an angular feedback with an encoder mounted on the motor shaft, while the load side encoder measures and evaluates the load angle, i.e. the displacement output from the gear transmission, connected to the load through a flexible coupling. The angular resolution of the load encoder is 0.45 arc-sec, while the apparent resolution of the motor encoder corresponds to 3.24 arc-sec. Inertia ratio between the actuator and the load is about 1:3 on the motor shaft.

III. MODELING OF ANGULAR TRANSMISSION ERROR

The angular transmission error θ_{TE} is generally defined by a motor angular displacement θ_M , a load angular displacement θ_L , and a gear ratio N as follows [4].

$$\theta_{TE} = \theta_L - \frac{\theta_M}{N} \quad (1)$$

TABLE I
SPECIFICATIONS OF PROTOTYPE.

gear ratio (N)	50
FS tooth number (z_f)	100
CS tooth number (z_c)	102
motor encoder resolution	8,000 pulse/rev.
load encoder resolution	2,880,000 pulse/rev.

In the following, in order to improve the static positioning accuracy at the settling, θ_{TE} is handled as the sum of the synchronous component θ_{Sync} and the nonlinear elastic component θ_{Hys} , as stated in the introduction.

A. Modeling for synchronous component

The synchronous component θ_{Sync} is basically caused by kinematic errors in teeth of FS and CS, and assembling errors in shaft of gearing and load, which is synchronous to the relative rotation of WG, FS, and CS [4]. According to the conventional studies, θ_{Sync} can be categorized into three components: θ_{TEM} ; synchronous component to motor angle, θ_{TEL} ; synchronous component to load angle, and θ_{TER} ; synchronous component to relative angle of FS and WG [14], [15]. In this research, however, only θ_{TEM} is modeled as $\theta_{Sync}(\theta_M)$, since 1) the load angle can not be detected for the positioning control, and 2) θ_{TEL} and θ_{TER} can be negligible in the amount comparing to θ_{TEM} . The synchronous component, therefore, is mathematically formulated as the following periodical pulsation for θ_M :

$$\theta_{Sync}(\theta_M) = \sum_{i=1}^{n_M} A_M(i) \cos(i\theta_M + \phi_M(i)), \quad (2)$$

where i : harmonic order for motor angle, A_M : amplitude of harmonics, and ϕ_M : phase of harmonics.

Fig.3 shows an experimental waveform and a spectrum analysis of the synchronous component. The transmission error waveform in the upper figure was indicated by the measured θ_M and θ_L for one revolution of load with a fine inching motion (step amplitude of 3.6 degrees on motor shaft) in clockwise (CW) direction. The solid lines in the lower figure are spectrum amplitudes of the experimental waveform for harmonic order. In the parameterization of $A_M(i)$ and $\phi_M(i)$, the dominant components in the spectrum amplitudes have been extracted, where larger components than a threshold indicated by a horizontal dashed line in the figure have been parameterized in eq.(2). The broken lines in the figure indicate the modeled spectrum amplitudes of harmonics, where parameters in the model are listed in Table II.

Fig.4 shows comparative waveforms of the synchronous component in the experiment (solid line) and the model (broken line), where the horizontal axis of load angle is magnified to scale three revolution of motor. From the figure, the model can precisely reproduce the actual synchronous component of transmission error.

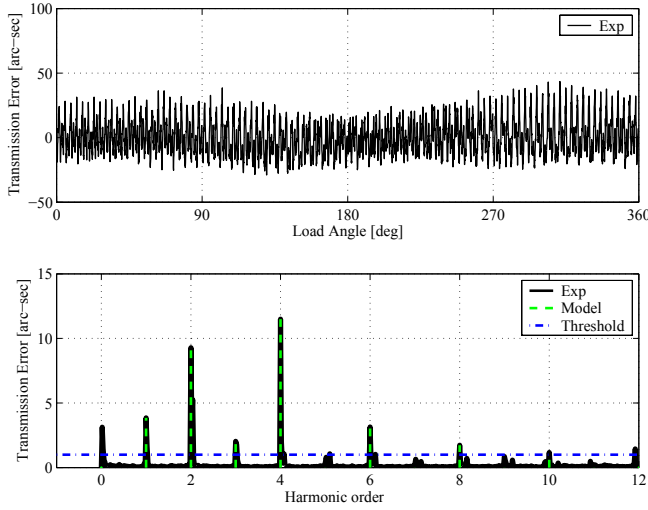


Fig. 3. Waveform and spectrum of synchronous component (CW).

TABLE II
PARAMETERS OF SYNCHRONOUS COMPONENT MODEL.

harmonic order i	$A_M(i)$ [arc-sec]	$\phi_M(i)$ [deg]
1	4.90	-126.8
2	9.33	-132.1
3	2.09	-96.2
4	12.21	105.7
6	3.18	-127.0
8	1.87	124.3
10	2.85	12.0

B. Modeling for nonlinear elastic component

In order to analyze and model the nonlinear elastic component, sinusoidal angular motions with amplitude of $180 \sim 5$ degrees on motor shaft have been driven, where an extra low frequency of 0.05 Hz was given to eliminate effects of inertia force and dynamic torsional vibration on the responses. Fig.5 shows waveforms of motor angle (upper) and transmission error (lower) for three sinusoidal angular references (blue: 180, red: 45, and green: 5 degrees, respectively). From the figure, the level of transmission error iterates between about 0 and 80 arc-sec at every velocity reversal point, i.e. at 10, 20, 30 s, while vibratory components due to the synchronous component are superimposed for the reference amplitude of 180 and 45 degrees.

Fig.6, on the other hand, indicates a Lissajous's figure of the transmission error for the motor angle in Fig.5, in order to clarify the level fluctuation in the error as a hysteresis behavior. The figure obviously suggests the following:

- the nonlinear elastic component of transmission error shows a hysteresis characteristic with the reproducibility, superimposing the periodical synchronous component for motor angle in large amplitude motions, and
- the inclination of the hysteresis curve at the velocity reversal is constant and independent of the sinusoidal amplitude.

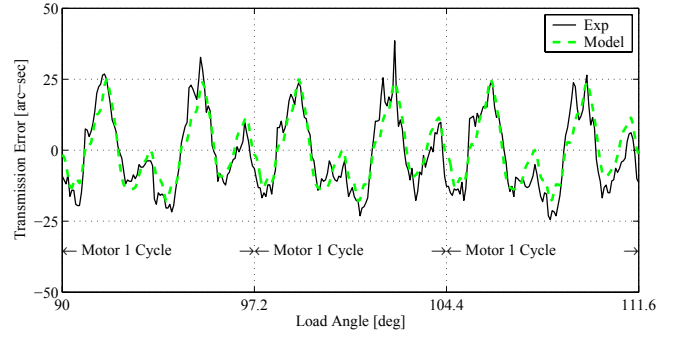


Fig. 4. Comparative waveforms of synchronous component in experiment and model.

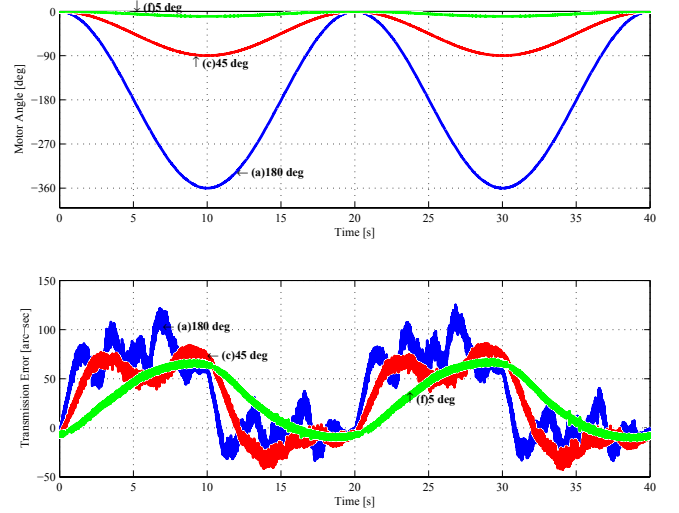


Fig. 5. Waveforms of angular transmission error for sinusoidal angular input.

Here, this nonlinear characteristic looks in common with the rolling friction behavior by elastic deformations and hysteresis attributes in micro-displacement regions of various motion devices. The nonlinear elastic component θ_{Hys} , therefore, is mathematically formulated by a hysteresis model with the nonlinear elastic deformation in FS, referring to a modeling framework of for the conventional rolling friction [11], [12], [13]. Fig.7 shows a conceptual nonlinear hysteresis model of θ_{Hys} for motor position θ_M , where the nonlinear elastic component can be characterized:

- 1) nonlinear elasticity varies depending on the motor angle θ_M ,
- 2) non-stable displacement region θ_r exists, where the error varies depending on the velocity direction, and
- 3) hysteresis behavior occurs in the non-stable region.

Based on these nonlinear behaviors, θ_{Hys} can be expressed as the following formula for the displacement δ after velocity

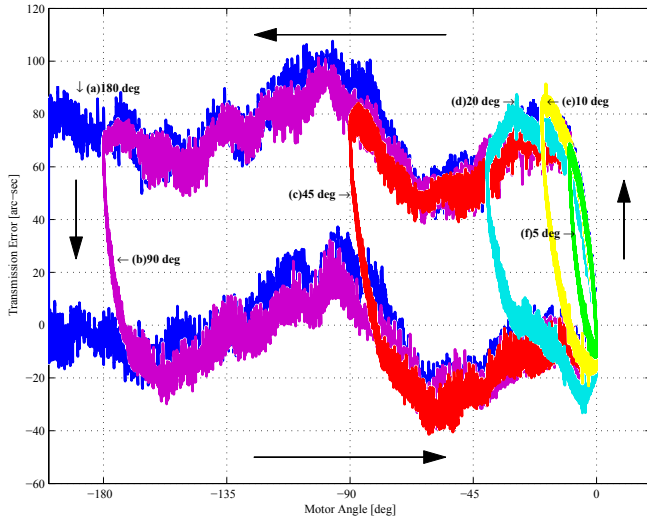


Fig. 6. Hysteresis characteristics of angular transmission error .

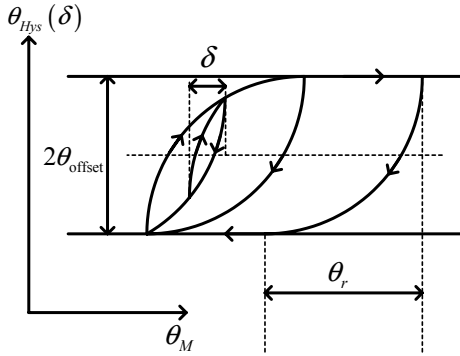


Fig. 7. Hysteresis model for nonlinear elastic deformation.

reversal:

$$\theta_{Hys}(\delta) = \begin{cases} \text{sgn}(\omega_M)(2\theta_{\text{offset}}g(\xi) - \theta'_{Hys}) & : |\delta| < \theta_r \text{ and } |\theta_{Hys}| < \theta_{\text{offset}} \\ \text{sgn}(\omega_M)\theta_{\text{offset}} & : |\delta| \geq \theta_r \text{ or } |\theta_{Hys}| \geq \theta_{\text{offset}} \end{cases} \quad (3)$$

$$g(\xi) = \begin{cases} \frac{1}{2-n}(\xi^{n-1} - (n-1)\xi) & : n \neq 2 \\ \xi(1 - \ln \xi) & : n = 2 \end{cases} \quad (4)$$

$$\delta = |\theta_M - \delta_0|, \xi = \delta/\theta_r \quad (5)$$

where ω_M is the motor angular velocity, $\text{sgn}(\omega_M)$ is the velocity sign function, n determines the width of hysteresis, θ'_{Hys} and δ_0 are the transmission error and the motor angle at the velocity reversal, and $2\theta_{\text{offset}}$ is the level fluctuation after the non-stable region.

The proposed model has been evaluated by comparing the response waveforms to the ones in experiments, where

TABLE III
PARAMETERS OF NONLINEAR ELASTIC MODEL.

θ_r	20 deg	n	1.6	θ_{offset}	40 arc-sec
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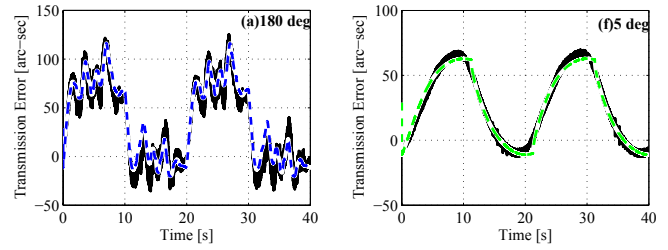


Fig. 8. Comparative response waveforms of angular transmission error in experiment and simulation.

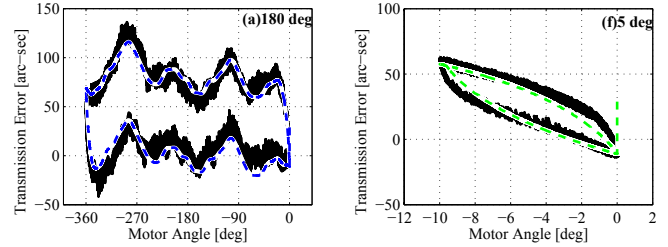


Fig. 9. Comparative hysteresis characteristics of angular transmission error in experiment and simulation.

the free parameters in the model was identified as listed in Table III with trial and error, to reproduce the hysteresis characteristics in Fig.6 subtracting the synchronous component. Figs.8 and 9 show comparative waveforms of angular transmission error and hysteresis characteristic in actual experiments (solid lines) and model-based simulations (dashed lines), where the same sinusoidal angular motions as of Fig.5 were driven with amplitude of 180 and 5 degrees on motor shaft. Notice here that the synchronous component is also superimposed in every waveform. From the figures, the reproducibility in the angular transmission error can be precisely achieved by the models for both the synchronous and nonlinear elastic models, verifying the effectiveness of the proposed mathematical modeling.

IV. EXPERIMENTAL RESULTS OF COMPENSATION FOR ANGULAR TRANSMISSION ERROR

The angular transmission error model has been adopted to the positioning system as a model-based feedforward compensation manner, in order to improve the steady state angular error due to the transmission error. Fig.10 shows a block diagram of the proposed angular transmission error compensation in a conventional positioning system, where the compensation angular input $N\theta_{TE}^* = N(\theta_{Sync} + \theta_{Hys})$ is subtracted from the original motor angular reference θ_M^* . In the following experimental verifications, inching motions of 240 steps (amplitude of 43.56 degrees in load side and an interval period of 2 s for every step) have been performed for both CW and CCW directions. Notice here that the inching step amplitude of 43.56 degrees was intentionally given in order to provide a different transmission error profile of the synchronous component in each inching motion. Feedback

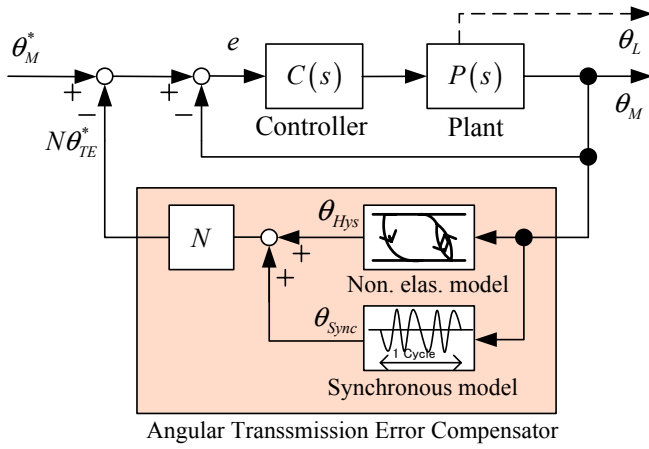


Fig. 10. Block diagram of angular transmission error compensator.

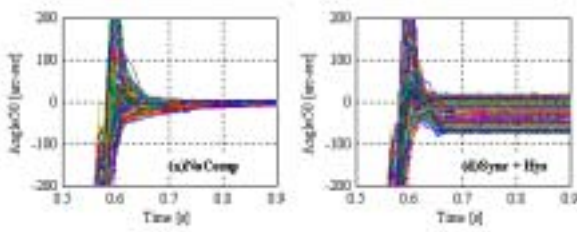


Fig. 11. Magnified response waveforms of motor angle at settling (CW).

controller $C(s)$ denotes a P-PI compensator (proportional for position, and proportional and integral for velocity).

Figs.11 and 12 show waveforms of the settling performance in motor and load angle responses for CW direction, where 240 inching trials are superimposed and the target angle is indicated as 0 arc-sec in each figure. In the figures, responses denote: (a) without transmission error compensation, (b) with compensation for only synchronous component θ_{Sync} , (c) with compensation for only nonlinear elastic component θ_{Hys} , and (d) with compensation for both θ_{Sync} and θ_{Hys} , respectively. From (a) of both Figs.11 and 12, the transmission error obviously causes steady state errors in load angle, while the motor can be settled at the exact target angle. Notice here that the scattered responses are observed during the transient in motor and load angles (overshoot and undershoot in responses), and at the settling in load angle (steady settling level), since the angular transmission error varies depending on the angle as analyzed in Section 3. In Fig.12 (d) with the compensation, on the other hand,

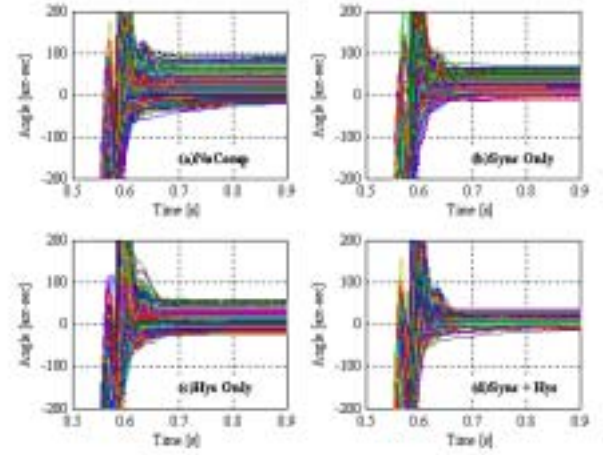


Fig. 12. Magnified response waveforms of load angle at settling (CW).

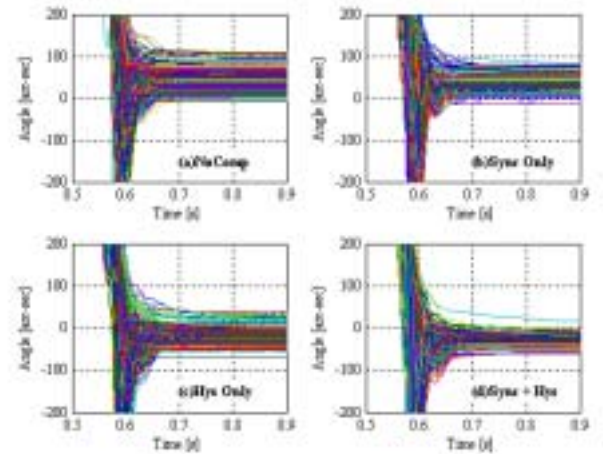


Fig. 13. Magnified response waveforms of load angle at settling (CCW).

the proposed compensation can suppress the scatter at the settling in load angle, as a result that the motor settling in Fig.11 (d) is shifting to the shorten target angle due to the angular reference compensation as shown in Fig.10. The applications of only each component in Fig.12 (b) and (c) are showing insufficiencies of the compensation, comparing to (d).

In order to quantitatively verify the compensation performance, statistical 3σ and mean values for the steady state error in load angle have been evaluated for each trial. Table IV lists a comparison of compensation performance with and without compensation, corresponding to Fig.12, where the first row in each value denotes the absolute value and the second row denotes the relative value which is normalized by the value without compensation. As the result of evaluation in 3σ values, the scattered settling accuracy can be successfully reduced to about 40% by the proposed compensation.

The same trials as of Fig.12 have been performed in CCW

TABLE IV

QUANTITATIVE COMPARISON OF COMPENSATION PERFORMANCE(CW).

	(a)	(b)	(c)	(d)	
3σ	75.4	51.9	52.8	28.1	arc-sec
	100.0	68.8	69.9	37.3	%
mean	19.4	25.8	11.2	12.3	arc-sec
	100.0	133.1	57.8	63.6	%

TABLE V
QUANTITATIVE COMPARISON OF COMPENSATION PERFORMANCE
(CCW).

	(a)	(b)	(c)	(d)	
3σ	75.0	50.2	55.7	30.2	arc-sec
	100	66.9	74.2	40.2	%
mean	34.2	29.0	-23.2	-29.15	arc-sec
	100.0	84.8	-67.7	-85.2	%

direction. Fig.13 shows response waveforms and Table V lists quantitative evaluations, respectively. In the CCW direction, the precise positioning performance can also achieved with the same accuracy as of in the CW direction.

V. CONCLUSIONS

The paper proposed a modeling and compensation methodology for the angular transmission error in harmonic drive gearings, especially paying attention to the performance improvement in static positioning accuracy at the settling. In the modeling, the angular transmission error has been classified into two components: the synchronous component and the nonlinear elastic component, according to its behaviors as a periodical pulsation and a nonlinearity with hysteresis attribution in micro displacement region. A novel mathematical formulation based on the modeling framework of the rolling friction with hysteresis, therefore, has been applied for the nonlinear elastic component, as well as the conventional modeling for the synchronous component by spectrum analyses for rotation angle, where the measurement apparatus and the parameterization procedure for the analysis and the modeling have been clearly stated in the paper. The proposed transmission error model has been adopted to the positioning system as a model-based feedforward compensation manner.

The numerical simulations and experiments using a laboratory positioning device have been performed to verify the proposed modeling and compensation. In the simulations, the mathematical model could precisely reproduce the transmission error with the synchronous and nonlinear elastic components. The positioning trials by an inching motion, on the other hand, the static positioning error could successfully reduce by the proposed compensation, where quantitatively verifications have been evaluated, i.e. about 40% of reduction of 3σ values in the scattered settling accuracy has been achieved.

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